39 A nucleus X is radioactive and decays into a nucleus Y.

X and Y are isotopes of the same element.

Which combination of particles could have been emitted during the decay process?

A 1 α-particle and 1 β^- particle

B 1 α-particle and 2 β^- particles

C 2 α-particles and 1 β^- particle

D 2 α-particles and 2 β^- particles